

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

SKILLS

• **Languages & DSA:** Python, SQL, Data Structures & Algorithms (Arrays, Hashing, Trees, Graphs, Dynamic Programming), LeetCode

• **Backend & Systems:** FastAPI, REST APIs, Microservices, Distributed Systems, ETL Pipelines, Kafka, PySpark

• **Databases, Cloud & Tools:** PostgreSQL, Neo4j, AWS (S3, EC2, EKS), GCP, Docker, Git

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

• Owned end-to-end design, development, and deployment of cross-platform software system for 100+ users, shipping high-quality features and continuously improving performance, reliability, and engineering standards

• Designed and optimized **PostgreSQL schema, indexing strategies, and query execution plans** for high-volume datasets, improving query performance by 38% under production workloads

• Built robust, production-ready **Python** pipeline with retrieval and context summarization, improving multi-turn response accuracy by **43%**

• Developed **backend workflows** for real-time, fault-tolerant processing of longitudinal health data with focus on reliability and scalability

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

• Designed and deployed highly scalable data processing pipelines in Python and Java to handle high-volume datasets, applying strong CS fundamentals to reduce processing latency by 41%

• Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing and pattern detection

• Designed and developed production-grade REST APIs using FastAPI and Node.js, leveraging AI-powered development tools to improve coding efficiency and problem-solving, with validation, error handling, and monitoring for high reliability

• Enabled large-scale data processing across social media streams using **distributed system architecture** principles

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

• Built and deployed scalable, microservices-based Retrieval-Augmented Generation (**RAG**) system using **FastAPI**, **PostgreSQL**, and vector embeddings for production-grade applications

• Designed efficient, fault-tolerant **ETL** pipelines to process and index large document datasets for low-latency, high-throughput retrieval systems

• Improved system performance by 64% through optimized indexing strategies and pipeline tuning, maintaining strong documentation and adhering to coding and architectural standards throughout production deployment

• Developed backend services enabling real-time, high-performance querying of unstructured data across distributed system architectures

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

• Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by **40%** and improving visibility into operational performance across transportation services and decision-making insights

• Processed and optimized large-scale operational datasets using **Python** and **PySpark** to identify failure patterns, trends, and performance issues in distributed systems environments

• Developed **SQL** and **graph queries** to map dependencies, analyze operations, detect data quality issues, speeding root-cause analysis by **26%**

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

• Developed and maintained **20+ Power BI** dashboards across Sales, Finance, and Operations for enterprise clients, leveraging **SQL** for data extraction and **DAX** for advanced calculations, enabling KPI tracking and data-driven decision-making

• Automated recurring **Excel**-based reporting workflows using **VBA**, implementing data validation, transformation, and scheduling logic, reducing report preparation time by **42%** and improving data consistency across weekly reporting cycles

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

• Led cross-functional team of 6 to ship product features rapidly using AI-powered development tools, leveraging performance insights to prioritize and deliver high-impact improvements in a fast-paced environment

• Tracked **sprint** progress, delivery metrics, and **QA issues** through **JIRA**-based workflows, managing backlog grooming, sprint planning, and issue resolution to improve cross-team visibility and ensure on-time milestone delivery

PROJECTS

Scalable ML System for Customer Risk Prediction

March 2025

UIUC Hackathon

• Built an **end-to-end machine learning system** using Python, Pandas, and LightGBM to forecast customer spending and assess credit risk from large-scale **time-series** transaction data, including data preprocessing, feature engineering, model training, and evaluation

• Developed a **decision framework for credit line adjustments**, incorporating model outputs and risk thresholds to improve prediction precision by 20% and enable risk-aware classification of customers